

BUS13-4

Microeconomics

Choice under constraint, prices, and the limits of markets

Every actor makes choices under constraint; prices coordinate, and markets have limits.

Albert School — 22 hours

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1 Foundations of economic reasoning

Economics studies how actors allocate scarce resources among competing uses. Because resources are finite, choosing one use forecloses others.

Definition 1 The **opportunity cost** of a choice is the value of the best alternative forgone in making it. It is the relevant cost of any decision, whether or not money changes hands.

Definition 2 **Marginal reasoning** evaluates a decision by its marginal benefit and marginal cost — the change in benefit and cost from one additional unit. An action is worth taking as long as its marginal benefit is at least its marginal cost.

Definition 3 A **sunk cost** is an expenditure already incurred that cannot be recovered. Sunk costs are economically irrelevant to current decisions: only future benefits and costs change with the choice now being made.

Definition 4 An **incentive** is a feature of the environment — a reward or penalty — that changes the marginal benefit or marginal cost of an action and therefore changes behaviour.

Definition 5 The **production possibility frontier** (PPF) is the set of maximum output combinations attainable from fixed resources and technology. Points inside the frontier are inefficient; points beyond it are unattainable. The slope of the frontier is the opportunity cost of one good in terms of the other.

Definition 6 A **positive** claim states what is — a description or prediction that can in principle be tested against evidence. A **normative** claim states what ought to be — a value judgement that cannot be settled by evidence alone.

Common pitfall Do not treat money already spent as a reason to continue. Only marginal, forward-looking benefits and costs are relevant; the sunk cost is gone under either choice.

2 Bounded rationality and behavioral realism

The **rationality hypothesis** assumes actors have consistent, well-ordered preferences and choose the option that best satisfies them given their constraints and information. It underlies the rational-choice claims used throughout the course. A body of behavioral findings challenges it.

Definition 7 Bounded rationality is the recognition that actors have limited information, limited attention, and limited computational capacity, and therefore settle for satisfactory rather than optimal choices.

Definition 8 Framing is the dependence of a choice on how options are described rather than on their substance: the same outcome presented as a gain or as a loss can yield different decisions.

Definition 9 Representativeness is the tendency to judge probability by similarity to a stereotype rather than by base rates, producing systematic errors of inference.

Definition 10 Loss aversion is the tendency to weigh losses more heavily than gains of equal size.

Definition 11 Present bias is the tendency to overweight immediate payoffs relative to future ones, producing inconsistent choices over time.

These findings bear on **incentive design**: incentives must account for how actors actually perceive and discount outcomes. This is distinct from marketing's consumer-psychology framing, which applies similar findings to persuasion rather than to the design of incentives.

3 Supply, demand, and market equilibrium

A market is described by a demand relation and a supply relation between price p and quantity q .

Definition 12 A demand curve gives the quantity buyers wish to purchase at each price, normally downward-sloping. A linear demand curve is written $q_d = a - bp$ with $a, b > 0$.

Definition 13 A supply curve gives the quantity sellers wish to sell at each price, normally upward-sloping. A linear supply curve is written $q_s = c + dp$ with $d > 0$.

Definition 14 Market equilibrium is the price p^* at which quantity demanded equals quantity supplied, $q_d = q_s$, and the corresponding quantity q^* . It is found by solving $a - bp = c + dp$, giving

$$p^* = \frac{a - c}{b + d}, \quad q^* = a - bp^*.$$

A **demand shock** shifts the demand curve (a change in a); a **supply shock** shifts the supply curve (a change in c). The new equilibrium is found by re-solving $q_d = q_s$ with the shifted intercept. An increase in demand raises both equilibrium price and quantity; an increase in supply lowers price and raises quantity.

4 Elasticities and welfare

Definition 15 The **price elasticity of demand** is the ratio of the percentage change in quantity demanded to the percentage change in price,

$$\varepsilon_p = \frac{dq}{dp} \cdot \frac{p}{q}.$$

Demand is **elastic** when $|\varepsilon_p| > 1$, **inelastic** when $|\varepsilon_p| < 1$, and **unit-elastic** when $|\varepsilon_p| = 1$.

Definition 16 The **income elasticity of demand** is $\varepsilon_I = \frac{dq}{dI} \cdot \frac{I}{q}$, where I is income. The good is **normal** when $\varepsilon_I > 0$ and **inferior** when $\varepsilon_I < 0$.

Definition 17 The **cross-price elasticity of demand** of good X with respect to the price of good Y is $\varepsilon_{XY} = \frac{dq_X}{dp_Y} \cdot \frac{p_Y}{q_X}$. The goods are **substitutes** when $\varepsilon_{XY} > 0$ and **complements** when $\varepsilon_{XY} < 0$.

The link between price elasticity and total revenue $R = pq$: a price increase raises revenue when demand is inelastic, lowers it when demand is elastic, and leaves it unchanged at unit elasticity.

Definition 18 **Consumer surplus** is the area between the demand curve and the market price up to the quantity traded: the buyers' total willingness to pay minus what they paid. **Producer surplus** is the area between the market price and the supply curve: revenue minus the sellers' minimum acceptable receipts.

Definition 19 **Deadweight loss** is the reduction in total surplus (consumer plus producer) caused by a departure from the competitive equilibrium quantity, such as a tax or market power.

5 Tax incidence

A **per-unit tax** t drives a wedge between the price paid by consumers p_c and the price received by producers p_s , with $p_c - p_s = t$. The post-tax quantity is found by solving $q_d(p_c) = q_s(p_s)$ subject to this wedge.

Definition 20 The **incidence** of a tax is the division of its burden between consumers and producers. The burden falls more heavily on the side of the market that is less elastic; the share borne by consumers is

$$\frac{d}{b+d}$$

for linear demand slope b and supply slope d (magnitudes).

The tax reduces consumer and producer surplus; part of the loss becomes government revenue tq and part becomes **deadweight loss** from the units no longer traded. The **real fiscal burden** on each side is its surplus loss net of any revenue returned, not the statutory side on which the tax is levied.

6 Consumer theory

Definition 21 A **utility function** $u(x, y)$ assigns to each bundle of goods a number representing the consumer's preference ordering. **Indifference curves** are the sets of bundles giving equal utility; their slope is the **marginal rate of substitution** (MRS), the rate at which the consumer will trade one good for the other while remaining indifferent.

Definition 22 The **budget constraint** is the set of affordable bundles, $p_x x + p_y y = I$, where I is income. Its slope is the price ratio $-\frac{p_x}{p_y}$.

Definition 23 The consumer's **optimal choice** is the affordable bundle on the highest attainable indifference curve, where the indifference curve is tangent to the budget line: $MRS = \frac{p_x}{p_y}$.

A price change is decomposed into two effects:

Definition 24 The **substitution effect** is the change in quantity demanded due to the change in relative prices, holding utility constant. The **income effect** is the change due to the change in real purchasing power. Their sum is the total response to the price change.

The business interpretation: the substitution effect captures switching toward relatively cheaper alternatives, while the income effect captures the change in overall purchasing power that a price movement causes.

7 Firm theory: production and costs

Definition 25 A **production function** $q = f(L, K)$ gives output from inputs (labour L , capital K). The **marginal product** of an input is the extra output from one more unit of it.

Definition 26 Diminishing returns: holding other inputs fixed, the marginal product of a variable input eventually falls as more of it is used.

Definition 27 Fixed cost (FC) does not vary with output; **variable cost (VC)** does. Total cost is $TC = FC + VC$. **Average total cost** is $ATC = \frac{TC}{q}$; **average variable cost** is $AVC = \frac{VC}{q}$. **Marginal cost** is the cost of one more unit, $MC = \frac{dTC}{dq}$.

Definition 28 The **break-even point** is the output at which price equals average total cost ($p = ATC$), so economic profit is zero. The **shutdown point** is the output at which price equals minimum average variable cost ($p = AVC$); below it the firm ceases production in the short run because revenue fails to cover variable cost.

8 Profit maximization

Definition 29 Marginal revenue (MR) is the extra revenue from selling one more unit. A firm maximizes profit at the output where **marginal cost equals marginal revenue**, $MC = MR$, provided price covers average variable cost.

Under perfect competition the firm is a price-taker, so $MR = p$ and the condition becomes $MC = p$.

Definition 30 The **zero-profit theorem**: under perfect competition with free entry and exit, economic profit tends to zero in the long run. Positive profit attracts entry, which lowers price; losses cause exit, which raises price; equilibrium is reached at $p = ATC_{\min}$.

9 Perfect and monopolistic competition

Definition 31 Perfect competition assumes many price-taking firms, a homogeneous product, and free entry. Each firm produces where $p = MC$; in the long run $p = ATC_{\min}$ and profit is zero.

Definition 32 Monopolistic competition assumes many firms selling differentiated products, each facing a downward-sloping demand curve, with free entry. In the short run a firm may earn profit by producing where $MC = MR$; entry of close substitutes erodes this until long-run profit is again zero, though output is below minimum-ATC scale.

10 Monopoly, price discrimination, and oligopoly

Definition 33 A **monopoly** is the sole seller in a market. Facing the whole downward-sloping demand curve, it sets output where $MC = MR$ and charges the price the demand curve allows at that output. Output is below, and price above, the competitive level, producing a deadweight loss.

To compute the monopoly outcome with linear demand $p = a - bq$: marginal revenue is $MR = a - 2bq$; set $MR = MC$ to find q_m , read p_m off the demand curve, and measure deadweight loss as the surplus lost between q_m and the competitive quantity where $p = MC$.

Definition 34 Price discrimination charges different prices for the same good to capture more surplus. **First-degree**: each unit sold at the buyer's maximum willingness to pay. **Second-degree**: price varies with quantity or version purchased (self-selection). **Third-degree**: different prices to identifiable groups with different elasticities.

Definition 35 An **oligopoly** is a market with few interdependent firms, so each firm's best action depends on the others'. Strategic interaction is represented by a **payoff matrix**.

Definition 36 A **Nash equilibrium** is a combination of strategies in which no player can gain by unilaterally changing strategy. In a 2×2 payoff matrix it is found by identifying each player's best response to each action of the other and locating the cell that is a mutual best response.

Definition 37 The **prisoner's dilemma** is a game in which each player has a dominant strategy to defect, yet mutual defection yields a worse outcome for both than mutual cooperation — illustrating how individually rational choices can be collectively inferior.

11 Market failures

Definition 38 An **externality** is a cost or benefit of a transaction borne by a third party, not reflected in the market price. A **negative externality** leads to over-production relative to the social optimum; a positive one, to under-production.

Definition 39 A **Pigouvian tax** is a per-unit tax equal to the marginal external cost at the socially optimal quantity. By raising private marginal cost to social marginal cost, it **internalizes** the externality and restores the efficient quantity.

Definition 40 **Asymmetric information** exists when one party to a transaction knows more than the other. **Adverse selection** arises before the transaction, when the informed party's hidden type drives the quality of who participates. **Moral hazard** arises after, when one party's hidden action changes once it is shielded from the consequences.

Definition 41 Akerlof's **market for lemons**: when buyers cannot distinguish high- from low-quality goods, they pay only an average price; sellers of high-quality goods withdraw, lowering average quality further, until the market unravels. It is the canonical case of adverse selection.

12 Bridge to macro measurement

Definition 42 Gross domestic product (GDP) is the market value of all final goods and services produced within a country in a period.

GDP can be measured by three equivalent approaches:

- **Production:** the sum of value added across all producers.
- **Expenditure:** $GDP = C + I + G + (X - M)$ — consumption, investment, government spending, and net exports.
- **Income:** the sum of incomes earned in production (wages, profits, rents, taxes net of subsidies).

Definition 43 GDP fails to capture significant phenomena, including non-market and household production, the informal economy, environmental degradation and resource depletion, the distribution of income, and changes in quality or leisure. It is a measure of recorded market activity, not of welfare.

This measurement literacy is what BUS23-4 Macroeconomics assumes when it opens on money, central banking, and the Keynesian multiplier.

Exercises

1. For a real resource-allocation decision, distinguish a positive from a normative claim about it, and apply opportunity cost and marginal reasoning, showing that a stated sunk cost is irrelevant.
2. Given linear demand $q_d = a - bp$ and supply $q_s = c + dp$, solve for equilibrium price and quantity, then recompute them after a stated supply or demand shock. Compute consumer surplus, producer surplus, and deadweight loss.
3. Compute the price, income, and cross-price elasticities of demand for a given market, and predict whether a proposed price change raises or reduces total revenue.
4. For a per-unit tax in a linear market, compute the post-tax prices paid and received, the surplus changes, the deadweight loss, and the share of the burden borne by consumers versus producers.
5. Given a utility function and a budget constraint, find the consumer's optimal bundle graphically, and decompose the response to a price change into income and substitution effects with a business interpretation.
6. State the rationality hypothesis and describe bounded rationality, framing, loss aversion, and present bias, giving for each an implication for incentive design distinct from a marketing application.
7. From production data, compute fixed, variable, marginal, and average costs, and identify the firm's break-even and shutdown points.
8. Apply $MC = MR$ numerically to find a firm's profit-maximizing output, and explain why economic profit tends to zero under perfect competition in the long run.
9. For a linear-demand monopoly, compute output, price, and deadweight loss; justify a first-, second-, and third-degree price-discrimination scheme; and solve for the Nash equilibrium in a 2×2 payoff matrix.

- 10.** Diagnose an externality, an adverse-selection situation via Akerlof's market for lemons, and a moral-hazard situation, and compute a Pigouvian tax that internalizes a given externality.
- 11.** Define GDP, explain the production, expenditure, and income approaches, and identify at least two significant phenomena GDP fails to capture.